Algorithm 1 PPO-Clip (Proximal Policy Optimization)

Ensure

Optimized policy $\pi_{\boldsymbol{\theta}}$

Hyperparameters

$\boldsymbol{\theta}_0$ — init policy params
$\boldsymbol{\phi}_0$ — init value params
ϵ — clip ratio (typ. 0.2)
γ — discount factor
λ — GAE param (0=TD, 1=MC)
N — actors
T — rollout length
K — epochs
M — minibatch size
α_π — policy learning rate
α_V — value learning rate
c_e — entropy coeff. (typ. 0.01)

Notation

s_t — state
a_t — action
r_t — reward
$V_{\boldsymbol{\phi}}(s)$ — value function
δ_t — TD error
$\hat{A}_t$ — GAE advantage
$\hat{R}_t$ — GAE-λ return
ρ_t — IS ratio $\pi_{\boldsymbol{\theta}}/\pi_{\boldsymbol{\theta}_k}$
$H[\cdot]$ — policy entropy

for $k = 0, 1, 2, \ldots$ **do**
  // Phase 1: Rollout collection
  for actor $= 1, \ldots, N$ **do**
    for $t = 0, \ldots, T{-}1$ **do**
      Sample action $a_t \sim \pi_{\boldsymbol{\theta}_k}(\cdot \mid s_t)$
      Execute a_t, observe r_t, s_{t+1}
      Store $(s_t, a_t, r_t, s_{t+1}, \pi_{\boldsymbol{\theta}_k}(a_t \mid s_t))$
    end for
  end for
  Collect buffer: $\mathcal{D}_k \leftarrow N \times T$ tuples
  // Phase 2: Advantage estimation (GAE) + normalization
  $S \leftarrow 0, \quad S_2 \leftarrow 0$
  for actor $= 1, \ldots, N$ **do**
    $\hat{A}_T \leftarrow 0$
    for $t = T{-}1, \ldots, 0$ **do**
      $\delta_t \leftarrow r_t + \gamma\, V_{\boldsymbol{\phi}_k}(s_{t+1}) - V_{\boldsymbol{\phi}_k}(s_t)$
      $\hat{A}_t \leftarrow \delta_t + \gamma\lambda\, \hat{A}_{t+1}$
      $\hat{R}_t \leftarrow \hat{A}_t + V_{\boldsymbol{\phi}_k}(s_t)$
      $(s_t, a_t, r_t, s_{t+1}, \pi_{\boldsymbol{\theta}_k}) \rightarrow (s_t, a_t, \hat{A}_t, \hat{R}_t, \pi_{\boldsymbol{\theta}_k})$
      $S \leftarrow S + \hat{A}_t, \quad S_2 \leftarrow S_2 + \hat{A}_t^2$
    end for
  end for
  $\mu_{\hat{A}} \leftarrow S/(N{\cdot}T), \quad \sigma_{\hat{A}} \leftarrow \sqrt{S_2/(N{\cdot}T) - \mu_{\hat{A}}^2}$
  for each $\hat{A}_t \in \mathcal{D}_k$ **do**
    $\hat{A}_t \leftarrow (\hat{A}_t - \mu_{\hat{A}})/(\sigma_{\hat{A}} + \varepsilon)$
  end for
  // Phase 3: Policy and value update
  Shuffle $\mathcal{D}_k$, split into $\lceil N{\cdot}T/M \rceil$ minibatches
  for epoch $= 1, \ldots, K$ **do**
    for each minibatch $\mathcal{B}$, $|\mathcal{B}| = M$ **do**
      $L^{\text{CLIP}} \leftarrow 0, \quad L^V \leftarrow 0, \quad H \leftarrow 0$
      for each $(s_t, a_t, \hat{A}_t, \hat{R}_t, \pi_{\boldsymbol{\theta}_k}) \in \mathcal{B}$ **do**
        $\rho_t \leftarrow \pi_{\boldsymbol{\theta}}(a_t \mid s_t)/\pi_{\boldsymbol{\theta}_k}(a_t \mid s_t)$
        $L^{\text{CLIP}} \mathrel{+}= \frac{1}{M} \min\big(\rho_t \hat{A}_t,\ \text{clip}(\rho_t, 1{-}\epsilon, 1{+}\epsilon)\hat{A}_t\big)$
        $L^V \mathrel{+}= \frac{1}{M}(V_{\boldsymbol{\phi}}(s_t) - \hat{R}_t)^2$
        for each action a in action space **do**
          $H \mathrel{-}= \frac{1}{M}\pi_{\boldsymbol{\theta}}(a \mid s_t) \log \pi_{\boldsymbol{\theta}}(a \mid s_t)$
        end for
      end for
      $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha_\pi \nabla_{\boldsymbol{\theta}}(L^{\text{CLIP}} + c_e H)$
      $\boldsymbol{\phi} \leftarrow \boldsymbol{\phi} - \alpha_V \nabla_{\boldsymbol{\phi}} L^V$
    end for
  end for
end for

Algorithm 2 PPO-Clip — Practical Vectorized Implementation

Ensure

Optimized policy $\pi_{\boldsymbol{\theta}}$

Hyperparameters

$\boldsymbol{\theta}_0$ — init policy params
$\boldsymbol{\phi}_0$ — init value params
ϵ — clip ratio (typ. 0.2)
γ — discount factor
λ — GAE param
N — parallel environments
T — rollout length
K — epochs
M — minibatch size
α_π — policy learning rate
α_V — value learning rate
c_e — entropy coeff.

Tensors

$\mathbf{S}, \mathbf{A}, \mathbf{R} \in \mathbb{R}^{N\times T}$
$\mathbf{D} \in \{0,1\}^{N\times T}$ — done flags
$\boldsymbol{\pi}_{\text{old}} \in \mathbb{R}^{N\times T}$
$\mathbf{V} \in \mathbb{R}^{N\times (T+1)}$
$\boldsymbol{\delta} \in \mathbb{R}^{N\times T}$
$\hat{\mathbf{A}}, \hat{\mathbf{R}} \in \mathbb{R}^{N\times T}$
$\boldsymbol{\rho} \in \mathbb{R}^{M}$

for $k = 0, 1, 2, \ldots$ **do**
  // Phase 1: Vectorized rollout $\mathcal{O}(T \cdot N \cdot C_\pi)$
  for $t = 0, \ldots, T{-}1$ **do**
    $\mathbf{A}_{:,t} \sim \pi_{\boldsymbol{\theta}_k}(\cdot \mid \mathbf{S}_{:,t})$ $\mathcal{O}(N \cdot C_\pi)$
    $\mathbf{R}_{:,t}, \mathbf{S}_{:,t+1}, \mathbf{D}_{:,t} \leftarrow \texttt{envs.step}(\mathbf{A}_{:,t})$ $\mathcal{O}(N)$
    $\boldsymbol{\pi}_{\text{old}_{:,t}} \leftarrow \pi_{\boldsymbol{\theta}_k}(\mathbf{A}_{:,t} \mid \mathbf{S}_{:,t})$
  end for
  // Phase 2: Vectorized GAE $\mathcal{O}(NT + N \cdot C_V)$
  $\mathbf{V} \leftarrow V_{\boldsymbol{\phi}_k}([\mathbf{S}, \mathbf{S}_{:,T}])$ $\mathcal{O}(N \cdot C_V)$; shape $(N, T{+}1)$
  $\boldsymbol{\delta} \leftarrow \mathbf{R} + \gamma\,(1 - \mathbf{D}) \odot \mathbf{V}_{:,1:} - \mathbf{V}_{:,:-1}$ $\mathcal{O}(NT)$
  $\hat{\mathbf{A}}_{:,T-1} \leftarrow \boldsymbol{\delta}_{:,T-1}$
  for $t = T{-}2, \ldots, 0$ **do**
    $\hat{\mathbf{A}}_{:,t} \leftarrow \boldsymbol{\delta}_{:,t} + \gamma\lambda\,(1 - \mathbf{D}_{:,t}) \odot \hat{\mathbf{A}}_{:,t+1}$ $\mathcal{O}(N)$
  end for
  $\hat{\mathbf{R}} \leftarrow \hat{\mathbf{A}} + \mathbf{V}_{:,:-1}$ $\mathcal{O}(NT)$
  // Phase 3: Minibatch SGD $\mathcal{O}(K \cdot NT \cdot (C_\pi + C_V))$
  Flatten $\mathcal{D}_k \leftarrow$ reshape all tensors to $(N{\cdot}T, \ldots)$
  for epoch $= 1, \ldots, K$ **do**
    $\mathcal{I} \leftarrow \texttt{randperm}(N{\cdot}T)$
    for $b = 0, M, 2M, \ldots$ **do** $\lceil NT/M \rceil$ minibatches
      $\mathcal{B} \leftarrow \mathcal{I}[b : b{+}M]$
      $\hat{\mathbf{A}}_{\mathcal{B}} \leftarrow (\hat{\mathbf{A}}_{\mathcal{B}} - \texttt{mean}(\hat{\mathbf{A}}_{\mathcal{B}}))/(\texttt{std}(\hat{\mathbf{A}}_{\mathcal{B}}) + \varepsilon)$ $\mathcal{O}(M)$
      $\boldsymbol{\rho} \leftarrow \pi_{\boldsymbol{\theta}}(\mathbf{A}_{\mathcal{B}} \mid \mathbf{S}_{\mathcal{B}}) \,/\, \boldsymbol{\pi}_{\text{old}_{\mathcal{B}}}$ $\mathcal{O}(M \cdot C_\pi)$
      $L^{\text{CLIP}} \leftarrow \texttt{mean}\big(\min(\boldsymbol{\rho} \odot \hat{\mathbf{A}}_{\mathcal{B}},\ \texttt{clip}(\boldsymbol{\rho}, 1{-}\epsilon, 1{+}\epsilon) \odot \hat{\mathbf{A}}_{\mathcal{B}})\big)$
      $L^{V} \leftarrow \texttt{mean}\big((V_{\boldsymbol{\phi}}(\mathbf{S}_{\mathcal{B}}) - \hat{\mathbf{R}}_{\mathcal{B}})^2\big)$ $\mathcal{O}(M \cdot C_V)$
      $H \leftarrow \texttt{mean}\big(\texttt{entropy}(\pi_{\boldsymbol{\theta}}(\cdot \mid \mathbf{S}_{\mathcal{B}}))\big)$
      $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha_\pi \nabla_{\boldsymbol{\theta}}(L^{\text{CLIP}} + c_e H)$ $\mathcal{O}(M \cdot C_\pi)$
      $\boldsymbol{\phi} \leftarrow \boldsymbol{\phi} - \alpha_V \nabla_{\boldsymbol{\phi}} L^{V}$ $\mathcal{O}(M \cdot C_V)$
    end for
  end for
end for

Per-iteration complexity: $\mathcal{O}\big(T{\cdot}N{\cdot}C_\pi + K{\cdot}N{\cdot}T{\cdot}(C_\pi + C_V)\big)$

C_π: policy forward/backward cost, C_V: value forward/backward cost.
Phase 3 dominates (K epochs $\times$ full buffer); Phase 2 is $\mathcal{O}(NT)$ arithmetic.

Algorithm 3 GRPO (Group Relative Policy Optimization)

Ensure

Optimized policy $\pi_{\boldsymbol{\theta}}$

Hyperparameters

$\boldsymbol{\theta}_0$ — init policy params
π_{ref} — frozen reference policy
ϵ — clip ratio (typ. 0.2)
G — group size
β — KL coefficient
α — learning rate
K — epochs
$R(\mathbf{q}, \mathbf{o})$ — reward function

Notation

$\mathbf{q}$ — input query (prompt)
$\mathbf{o}_j$ — j-th completion
$o_{j,t}$ — token t of completion j
$\mathbf{o}_{j,<t}$ — preceding tokens
$|\mathbf{o}_j|$ — completion length
$\hat{A}_j$ — group-relative advantage
$\rho_{j,t}$ — IS ratio $\pi_{\boldsymbol{\theta}}/\pi_{\boldsymbol{\theta}_k}$
D_{KL} — KL divergence

for $k = 0, 1, 2, \ldots$ **do**
  // Phase 1: Group sampling
  for each query $\mathbf{q}$ in training batch **do**
    for $j = 1, \ldots, G$ **do**
      Sample completion $\mathbf{o}_j \sim \pi_{\boldsymbol{\theta}_k}(\cdot \mid \mathbf{q})$
      Compute reward $R_j \leftarrow R(\mathbf{q}, \mathbf{o}_j)$
      for $t = 1, \ldots, |\mathbf{o}_j|$ **do**
        Cache $\pi_{\boldsymbol{\theta}_k}(o_{j,t} \mid \mathbf{q},\, \mathbf{o}_{j,<t})$
      end for
    end for
  end for
  // Phase 2: Group-relative advantage (no critic)
  for each query group $\{R_1, \ldots, R_G\}$ **do**
    $\mu \leftarrow 0, \quad \sigma^2 \leftarrow 0$
    for $j = 1, \ldots, G$ **do**
      $\mu \leftarrow \mu + R_j/G$
    end for
    for $j = 1, \ldots, G$ **do**
      $\sigma^2 \leftarrow \sigma^2 + (R_j - \mu)^2/G$
    end for
    for $j = 1, \ldots, G$ **do**
      $\hat{A}_j \leftarrow (R_j - \mu)/\max(\sqrt{\sigma^2},\, \varepsilon_{\min})$
    end for
  end for
  // Phase 3: Clipped surrogate update
  for epoch $= 1, \ldots, K$ **do**
    for each minibatch $\mathcal{B}$ of completions **do**
      $L^{\text{GRPO}} \leftarrow 0, \quad D_{\text{KL}} \leftarrow 0$
      for each $(\mathbf{q}, \mathbf{o}_j, \hat{A}_j) \in \mathcal{B}$ **do**
        for $t = 1, \ldots, |\mathbf{o}_j|$ **do**
          $\rho_{j,t} \leftarrow \pi_{\boldsymbol{\theta}}(o_{j,t} \mid \mathbf{q}, \mathbf{o}_{j,<t}) / \pi_{\boldsymbol{\theta}_k}(o_{j,t} \mid \mathbf{q}, \mathbf{o}_{j,<t})$
          $L^{\text{GRPO}} \mathrel{+}= \frac{1}{|\mathcal{B}||\mathbf{o}_j|} \min\big(\rho_{j,t}\hat{A}_j,\ \text{clip}(\rho_{j,t}, 1-\epsilon, 1+\epsilon)\hat{A}_j\big)$
          $D_{\text{KL}} \mathrel{+}= \frac{1}{|\mathcal{B}|}\big[\frac{\pi_{\boldsymbol{\theta}}}{\pi_{\text{ref}}} - \log\frac{\pi_{\boldsymbol{\theta}}}{\pi_{\text{ref}}} - 1\big]$
        end for
      end for
      $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha \nabla_{\boldsymbol{\theta}}(L^{\text{GRPO}} - \beta\, D_{\text{KL}})$
    end for
  end for
end for

Algorithm 4 CISPO (Clipped Importance Sampling Policy Optimization)

Ensure

Optimized policy $\pi_{\boldsymbol{\theta}}$

Hyperparameters

$\boldsymbol{\theta}_0$ — init policy params
ϵ_{high} — upper clip (typ. 0.3)
G — group size
α — learning rate
K — epochs
$R(\mathbf{q}, \mathbf{o})$ — reward function

Notation

Same as GRPO, plus:
$\hat{\rho}_{j,t}$ — clamped IS ratio
$\bar{\rho}_{j,t}$ — detached weight
$\textsc{StopGrad}(\cdot)$ — detach from computation graph
N_{tokens} — total tokens in batch

for $k = 0, 1, 2, \ldots$ **do**
  // Phase 1: Group sampling (same as GRPO)
  for each query $\mathbf{q}$ in training batch **do**
    for $j = 1, \ldots, G$ **do**
      Sample completion $\mathbf{o}_j \sim \pi_{\boldsymbol{\theta}_k}(\cdot \mid \mathbf{q})$
      Compute reward $R_j \leftarrow R(\mathbf{q}, \mathbf{o}_j)$
      for $t = 1, \ldots, |\mathbf{o}_j|$ **do**
        Cache $\pi_{\boldsymbol{\theta}_k}(o_{j,t} \mid \mathbf{q},\ \mathbf{o}_{j,<t})$
      end for
    end for
  end for
  // Phase 2: Group-relative advantage (same as GRPO)
  for each query group $\{R_1, \ldots, R_G\}$ **do**
    $\mu \leftarrow 0, \quad \sigma^2 \leftarrow 0$
    for $j = 1, \ldots, G$ **do**
      $\mu \leftarrow \mu + R_j / G$
    end for
    for $j = 1, \ldots, G$ **do**
      $\sigma^2 \leftarrow \sigma^2 + (R_j - \mu)^2 / G$
    end for
    for $j = 1, \ldots, G$ **do**
      $\hat{A}_j \leftarrow (R_j - \mu) / \max(\sqrt{\sigma^2},\ \varepsilon_{\min})$
    end for
  end for
  // Phase 3: Clipped IS weight update
  // Key: clip weight + detach → all tokens get gradient
  for epoch $= 1, \ldots, K$ **do**
    for each minibatch $\mathcal{B}$ of completions **do**
      $L^{\text{CISPO}} \leftarrow 0, \quad N_{\text{tokens}} \leftarrow 0$
      for each $(\mathbf{q}, \mathbf{o}_j, \hat{A}_j) \in \mathcal{B}$ **do**
        $N_{\text{tokens}} \leftarrow N_{\text{tokens}} + |\mathbf{o}_j|$
        for $t = 1, \ldots, |\mathbf{o}_j|$ **do**
          $\rho_{j,t} \leftarrow \pi_{\boldsymbol{\theta}}(o_{j,t} \mid \mathbf{q}, \mathbf{o}_{j,<t}) / \pi_{\boldsymbol{\theta}_k}(o_{j,t} \mid \mathbf{q}, \mathbf{o}_{j,<t})$
          $\hat{\rho}_{j,t} \leftarrow \text{clamp}(\rho_{j,t},\ \max = 1 + \epsilon_{\text{high}})$
          $\bar{\rho}_{j,t} \leftarrow \textsc{StopGrad}(\hat{\rho}_{j,t})$
          $L^{\text{CISPO}} \mathrel{+}= \bar{\rho}_{j,t} \cdot \hat{A}_j \cdot \log \pi_{\boldsymbol{\theta}}(o_{j,t} \mid \mathbf{q}, \mathbf{o}_{j,<t})$
        end for
      end for
      $L^{\text{CISPO}} \leftarrow L^{\text{CISPO}} / N_{\text{tokens}}$
      $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha \nabla_{\boldsymbol{\theta}} L^{\text{CISPO}}$
    end for
  end for
end for

Algorithm 5 GRPO — Practical Vectorized Implementation

Ensure

Optimized policy $\pi_{\boldsymbol{\theta}}$

Hyperparameters

$\boldsymbol{\theta}_0$ — init policy params
π_{ref} — frozen ref. policy
ϵ — clip ratio (typ. 0.2)
G — group size
β — KL coefficient
α — learning rate
K — epochs
M — minibatch size
$R(\mathbf{q}, \mathbf{o})$ — reward function

Tensors

B — queries per batch
L — max completion length
$\mathbf{Q} \in \mathbb{Z}^{B \times L_q}$ — queries
$\mathbf{O} \in \mathbb{Z}^{B \times G \times L}$ — tokens
$\boldsymbol{\pi}_{\text{old}} \in \mathbb{R}^{B \times G \times L}$
$\mathbf{M} \in \{0,1\}^{B \times G \times L}$ — mask
$\mathbf{R} \in \mathbb{R}^{B \times G}$ — rewards
$\hat{\mathbf{A}} \in \mathbb{R}^{B \times G}$ — advantages
$\boldsymbol{\rho} \in \mathbb{R}^{M \times L}$

for $k = 0, 1, 2, \ldots$ **do**
  // Phase 1: Batch generation $\mathcal{O}(B{\cdot}G{\cdot}L{\cdot}C_\pi)$
  for each query batch $\mathbf{Q}_{1:B}$ **do**
    $\mathbf{O}_{:,:,:} \sim \pi_{\boldsymbol{\theta}_k}(\cdot \mid \mathbf{Q})$ $B{\cdot}G$ completions in parallel
    $\mathbf{R}_{:,:} \leftarrow R(\mathbf{Q}, \mathbf{O})$ batch reward evaluation
    $\boldsymbol{\pi}_{\text{old}} \leftarrow \pi_{\boldsymbol{\theta}_k}(\mathbf{O} \mid \mathbf{Q})$ $\mathcal{O}(BGL{\cdot}C_\pi)$
    $\mathbf{M} \leftarrow \texttt{token_mask}(\mathbf{O})$ 1 where valid, 0 for padding
  end for
  // Phase 2: Group-relative advantage $\mathcal{O}(BG)$
  $\boldsymbol{\mu} \leftarrow \texttt{mean}(\mathbf{R}, \dim = G)$ $\in \mathbb{R}^B$
  $\boldsymbol{\sigma} \leftarrow \texttt{std}(\mathbf{R}, \dim = G)$ $\in \mathbb{R}^B$
  $\hat{\mathbf{A}} \leftarrow (\mathbf{R} - \boldsymbol{\mu}) \,/\, \max(\boldsymbol{\sigma}, \varepsilon_{\min})$ broadcast over G
  // Phase 3: Minibatch SGD $\mathcal{O}(K{\cdot}BG{\cdot}L{\cdot}C_\pi)$
  Flatten $\mathcal{D}_k \leftarrow$ reshape to $(B{\cdot}G, L)$ completions
  for epoch $= 1, \ldots, K$ **do**
    $\mathcal{I} \leftarrow \texttt{randperm}(B{\cdot}G)$ shuffle completions
    for $b = 0, M, 2M, \ldots$ **do** $\lceil BG/M \rceil$ minibatches
      $\mathcal{B} \leftarrow \mathcal{I}[b : b{+}M]$
      $\boldsymbol{\rho} \leftarrow \pi_{\boldsymbol{\theta}}(\mathbf{O}_{\mathcal{B}} \mid \mathbf{Q}_{\mathcal{B}}) \,/\, \boldsymbol{\pi}_{\text{old}_{\mathcal{B}}}$ $\mathcal{O}(ML{\cdot}C_\pi)$
      $\mathbf{L}_{\text{tok}} \leftarrow \min\big(\boldsymbol{\rho} \odot \hat{\mathbf{A}}_{\mathcal{B}},\ \texttt{clip}(\boldsymbol{\rho}, 1{-}\epsilon, 1{+}\epsilon) \odot \hat{\mathbf{A}}_{\mathcal{B}}\big)$
      $L^{\text{GRPO}} \leftarrow \texttt{sum}(\mathbf{L}_{\text{tok}} \odot \mathbf{M}_{\mathcal{B}}) \,/\, \texttt{sum}(\mathbf{M}_{\mathcal{B}})$ masked mean
      $\mathbf{r}_{\text{ref}} \leftarrow \pi_{\boldsymbol{\theta}}(\mathbf{O}_{\mathcal{B}} \mid \cdot) \,/\, \pi_{\text{ref}}(\mathbf{O}_{\mathcal{B}} \mid \cdot)$
      $D_{\text{KL}} \leftarrow \texttt{mean}\big((\mathbf{r}_{\text{ref}} - \log \mathbf{r}_{\text{ref}} - 1) \odot \mathbf{M}_{\mathcal{B}}\big)$
      $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha \nabla_{\boldsymbol{\theta}}(L^{\text{GRPO}} - \beta\, D_{\text{KL}})$ $\mathcal{O}(ML{\cdot}C_\pi)$
    end for
  end for
end for

Per-iteration complexity: $\mathcal{O}\big(B{\cdot}G{\cdot}L{\cdot}C_\pi + K{\cdot}B{\cdot}G{\cdot}L{\cdot}C_\pi\big) = \mathcal{O}\big((1{+}K){\cdot}BGL{\cdot}C_\pi\big)$

C_π: policy forward/backward cost. No C_V term (critic-free).
Phase 2 is $\mathcal{O}(BG)$ arithmetic — negligible vs. neural forward passes.

Algorithm 6 CISPO — Practical Vectorized Implementation

Ensure

Optimized policy $\pi_{\boldsymbol{\theta}}$

Hyperparameters

$\boldsymbol{\theta}_0$ — init policy params
ϵ_{high} — upper clip (typ. 0.3)
G — group size
α — learning rate
K — epochs
M — minibatch size
$R(\mathbf{q}, \mathbf{o})$ — reward function

Tensors

B — queries per batch
L — max completion length
$\mathbf{Q} \in \mathbb{Z}^{B \times L_q}$ — queries
$\mathbf{O} \in \mathbb{Z}^{B \times G \times L}$ — tokens
$\boldsymbol{\pi}_{\text{old}} \in \mathbb{R}^{B \times G \times L}$
$\mathbf{M} \in \{0,1\}^{B \times G \times L}$ — mask
$\mathbf{R} \in \mathbb{R}^{B \times G}$ — rewards
$\hat{\mathbf{A}} \in \mathbb{R}^{B \times G}$ — advantages
$\boldsymbol{\rho}, \hat{\boldsymbol{\rho}}, \bar{\boldsymbol{\rho}} \in \mathbb{R}^{M \times L}$

for $k = 0, 1, 2, \ldots$ **do**
  // Phase 1: Batch generation (same as GRPO) $\mathcal{O}(BGL{\cdot}C_\pi)$
  for each query batch $\mathbf{Q}_{1:B}$ **do**
    $\mathbf{O}_{:,:,:} \sim \pi_{\boldsymbol{\theta}_k}(\cdot \mid \mathbf{Q})$ — $B{\cdot}G$ completions in parallel
    $\mathbf{R}_{:,:} \leftarrow R(\mathbf{Q}, \mathbf{O})$ — batch reward evaluation
    $\boldsymbol{\pi}_{\text{old}} \leftarrow \pi_{\boldsymbol{\theta}_k}(\mathbf{O} \mid \mathbf{Q})$ — $\mathcal{O}(BGL{\cdot}C_\pi)$
    $\mathbf{M} \leftarrow \texttt{token_mask}(\mathbf{O})$ — 1 where valid, 0 for padding
  end for
  // Phase 2: Group-relative advantage (same as GRPO) $\mathcal{O}(BG)$
  $\boldsymbol{\mu} \leftarrow \texttt{mean}(\mathbf{R}, \dim = G)$ — $\in \mathbb{R}^B$
  $\boldsymbol{\sigma} \leftarrow \texttt{std}(\mathbf{R}, \dim = G)$ — $\in \mathbb{R}^B$
  $\hat{\mathbf{A}} \leftarrow (\mathbf{R} - \boldsymbol{\mu}) \,/\, \max(\boldsymbol{\sigma}, \varepsilon_{\min})$ — broadcast over G
  // Phase 3: Clipped IS weight SGD $\mathcal{O}(K{\cdot}BG{\cdot}L{\cdot}C_\pi)$
  Flatten $\mathcal{D}_k \leftarrow$ reshape to $(B{\cdot}G, L)$ completions
  for epoch $= 1, \ldots, K$ **do**
    $\mathcal{I} \leftarrow \texttt{randperm}(B{\cdot}G)$
    for $b = 0, M, 2M, \ldots$ **do** — $\lceil BG/M \rceil$ minibatches
      $\mathcal{B} \leftarrow \mathcal{I}[b : b{+}M]$
      $\boldsymbol{\rho} \leftarrow \pi_{\boldsymbol{\theta}}(\mathbf{O}_{\mathcal{B}} \mid \mathbf{Q}_{\mathcal{B}}) \,/\, \boldsymbol{\pi}_{\text{old}_{\mathcal{B}}}$ — $\mathcal{O}(ML{\cdot}C_\pi)$
      $\hat{\boldsymbol{\rho}} \leftarrow \texttt{clamp}(\boldsymbol{\rho},\ \max = 1 + \epsilon_{\text{high}})$ — $\mathcal{O}(ML)$
      $\bar{\boldsymbol{\rho}} \leftarrow \texttt{stop_grad}(\hat{\boldsymbol{\rho}})$ — detach from graph
      $\mathbf{L}_{\text{tok}} \leftarrow \bar{\boldsymbol{\rho}} \odot \hat{\mathbf{A}}_{\mathcal{B}} \odot \log \pi_{\boldsymbol{\theta}}(\mathbf{O}_{\mathcal{B}} \mid \mathbf{Q}_{\mathcal{B}})$
      $L^{\text{CISPO}} \leftarrow \texttt{sum}(\mathbf{L}_{\text{tok}} \odot \mathbf{M}_{\mathcal{B}}) \,/\, \texttt{sum}(\mathbf{M}_{\mathcal{B}})$ — masked mean
      $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha \nabla_{\boldsymbol{\theta}} L^{\text{CISPO}}$ — $\mathcal{O}(ML{\cdot}C_\pi)$
    end for
  end for
end for

Per-iteration complexity: $\mathcal{O}\big((1{+}K){\cdot}BGL{\cdot}C_\pi\big)$ — same as GRPO

No π_{ref} forward pass needed (no KL term). `stop_grad` is $\mathcal{O}(1)$.
All tokens receive gradient (vs. GRPO where clipped tokens are masked out).

Algorithm 7 Key Differences: PPO vs GRPO vs CISPO

PPO *(Schulman et al., 2017):*
$L^{\text{CLIP}} = \min\big(\rho_t \hat{A}_t,\ \ \text{clip}(\rho_t,\ 1{-}\epsilon,\ 1{+}\epsilon)\, \hat{A}_t\big)$ (ascent)
Advantage via *learned critic* V_ϕ.
When $\rho_t > 1{+}\epsilon$ and $\hat{A}_t > 0$: $\nabla_{\boldsymbol{\theta}} = \mathbf{0}$ *(token masked out).*

GRPO *(DeepSeek, 2024):*
$L = -\min\big(\rho_{j,t} \hat{A}_j,\ \ \text{clip}(\rho_{j,t},\ 1{-}\epsilon,\ 1{+}\epsilon)\, \hat{A}_j\big)$
Advantage via *group-relative normalization* (no critic).
When $\rho_{j,t} > 1{+}\epsilon$ and $\hat{A}_j > 0$: $\nabla_{\boldsymbol{\theta}} = \mathbf{0}$ *(token masked out).*

CISPO *(MiniMax-M1, 2025):*
$L = -\text{SG}\big(\text{clamp}(\rho_{j,t},\ \max{=}1{+}\epsilon_h)\big) \cdot \hat{A}_j \cdot \log \pi_\theta$
Advantage via *group-relative normalization* (no critic).
For all tokens: $\nabla_{\boldsymbol{\theta}} \neq \mathbf{0}$ *(gradient always flows through* $\log \pi$*).*